

Becoming: An Interactive Musical Journey in VR

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ABSTRACT

Becoming is an operatic VR piece based on a Persian poem by Mowlana Rumi. The critical content of the piece is about the spiritual evolution of humans on Earth. The artistic expression of the piece takes advantage of an advanced ray-tracing audio spatialization system (Space3D), which is capable of creating realistic spatial impressions within changing acoustic environments in real-time. In this piece the user can interact with the environment and influence the progression of the music by touching various elements and by changing the spatialization paths and speeds of various layers of the music. Two audience members can be connected through the network and interact with each other via haptic effects.

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1 INTRODUCTION

In traditional theatrical and cinematic contexts, the proscenium and camera angles as well as lighting guide the formal focal points for the audience members. The immersive aspects of VR and XR applications provide a greater freedom for the audience to choose their point of focus. In this context, because of the immersive perceptual cues of our auditory system, the role of audio and music becomes more important in providing narrative focal cues and defining the aural properties of the immersive world. The use of the Space3D system and haptics afford *Becoming* a fully immersive context for delivery of spatial and interactive music to multiple

users connected through the network. In this piece we present a context for two participants, but the model can be extended to any number of participants experiencing the same virtual environment.

2 THE MUSIC

The musical form relies on the interaction among multiple delayed tracks of the same melodic and percussive lines. This delay processing was used at the time of all recordings (vocals, strings, and percussion) and thus the performers have musically reacted to the emerging forms. Synthetic sounds have been used throughout the piece, at times in supporting the calm sense of the acoustic malarial and at times for supporting the critical narrative. The relationship among the delayed musical lines is used as a vehicle of interactivity for the participants to change or add to the mix of the music. Additional musical tracks are attached to the controllers of both remote and local participants. The scenes can be aurally animated by activating the controllers as sonic objects and moving them within the communal world of the two participants. During this piece many musical tracks (at times up to 25 tracks) are being spatialized at different distances and in varying paths around the participants.

3 SPATIALIZATION AND INTERACTIVITY

Multimodality is a key principle in the design of the interactive components. One example of this is a dynamic sound source engine based on a flocking algorithm. In this system, a set of sound sources are mapped to a set of visual assets that are entwined in a flocking system to determine their movement, where the positions of these assets are mirrored by the spatialization of the sounds. The visual assets themselves are simple particle systems with parameters that modulate according to the audio streams. This is made interactive for the user by adding the sum of the velocity vectors of the hands to the velocities of each sound source, allowing for dramatic gestures like waving the sources away from the user.

The audio in *Becoming* is rendered through Space3D, a real-time audio spatialization and acoustic modeling system developed by Sonic Arts Research and Development group at Qualcomm Institute. The system produces convincing, high-quality spatial impressions for a human listener over speakers, headphones, or both

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simultaneously. Space3D uses NVIDIA RTX real-time ray tracing hardware to find acoustical paths through the scene, and processes the audio traversing those paths with CUDA. The system models high-order specular reflections, shadowed and non-shadowed diffraction (including on reflection paths), per-vertex material reflection impulse response and transmission coefficients, source directionality, listener HRTFs, Doppler, and more. The spatialization algorithm is an evolution of a technique proposed by F. Richard Moore [Moore 1983; Yadegari 2005]. The algorithm has been considerably extended and adapted for 3D in the context of real-time geometrical acoustics.

In Space3D, all geometrical and audio processing is performed on a per-frame basis, needing no precomputed information about the scene, so Space3D can react to arbitrary scene changes such as deforming meshes. Furthermore, Space3D does not merely create a room impulse response every frame and crossfade between them as the scene changes—it simulates the parameters of each sound path, such as path length and filtering, as they change continuously over the frame. This is needed for correct simulation of changing delays for the speaker-based spatialization algorithm, as well as Doppler and generally higher-quality results. Space3D simulates diffraction and transmission using the VDaT algorithm [Pisha et al. 2020], which traces rays to sample the space around the original path, finds candidate paths around obstacles, and uses these results to approximate the high-quality BTM edge diffraction model [Svensson and Calamia 2006] at a much lower computational cost.

4 VISUAL DESIGN

The visual design of *Becoming* includes interactive visual effects which comprise each of the various worlds. In designing the visual components of the work, the main emphasis was on imagining how discrete environments could smoothly transform into other elements, crafting a visual narrative from that series of interpolations.

One element of the piece which exemplifies this approach is the system which underlies the water-to-mountain transformation. This asset consists of a single high-resolution plane which crossfades between two processes. The first is a stock procedural water material that interactively computes vertex and normal displacement based off of collisions. The second is a custom Fractional Brownian Motion (FBM) terrain displacement shader [Echelard et al. 2010] which computes vertex and normal displacement from an underlying dynamic noise generator, with parameters that can be modulated in real-time. Shifting from the water to land is then a simple process of crossfading between these generators in the fragment and vertex shaders.

The world of *Becoming* is a continuously evolving ecological, and ultimately industrial, narrative depicting various stages of life and being. In the midst of this process, the human emerges at the pivotal moment of creation turning into destruction. The treatment of the human form as a pointcloud allows for it to be treated as amorously as the environments which precede—and coincide—with it.

5 HAPTICS

A set of full-body wearable haptic devices with a high spatial resolution (bHaptics, TactSuit X40 and TacTosy for Arms) are used

for presenting haptic sensations to the users to complement the audiovisual experiences [Melo et al. 2022]. The benefits of this approach have been demonstrated in many VR applications [Covaci et al. 2018]. On interacting with visual assets in *Becoming*, tactile feedback is provided to improve the sense of presence. At important scenic moments, feedback is used to add another layer of experience supporting the other senses. While interaction-based tactile effects are localized to the site of the interaction, scenic effects spread across the arms and torso with patterns reflecting what occurs around the audience member. For example, spatialized sounds are at some points used to dynamically control moving vibration. As tactile feedback alone can only partly replicate real-world phenomena, like touching a plant, the effects take on a metaphorical role to heighten one's exploration of the changing world rather than replicate familiar sensations. Similarly, feedback is used to enhance the relationship between the local and remote participants. Effects are felt when the participants shake hands or touch the body of the other participant.

6 SUMMARY

In this piece we have experimented with the use of a high quality audio spatializer (Space3D) and haptic effects to build a multimodal interactive musical artifact. One of the applications of Space3D is delivery of a form of music where audience members can choose their location of listening in relation to performers and instruments, and the acoustic environment in which they listen to music. The piece allows two users to experience and modify the same visual and aural environment. Users experience haptic feedback both in touching objects in the world, or in touching each other's virtual body parts.

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